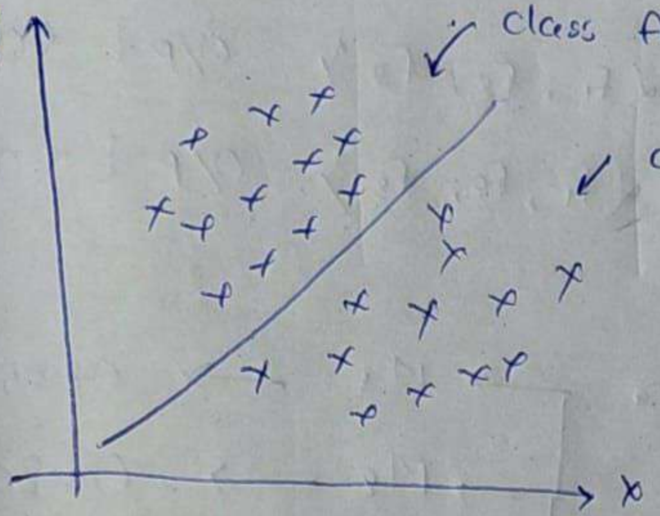
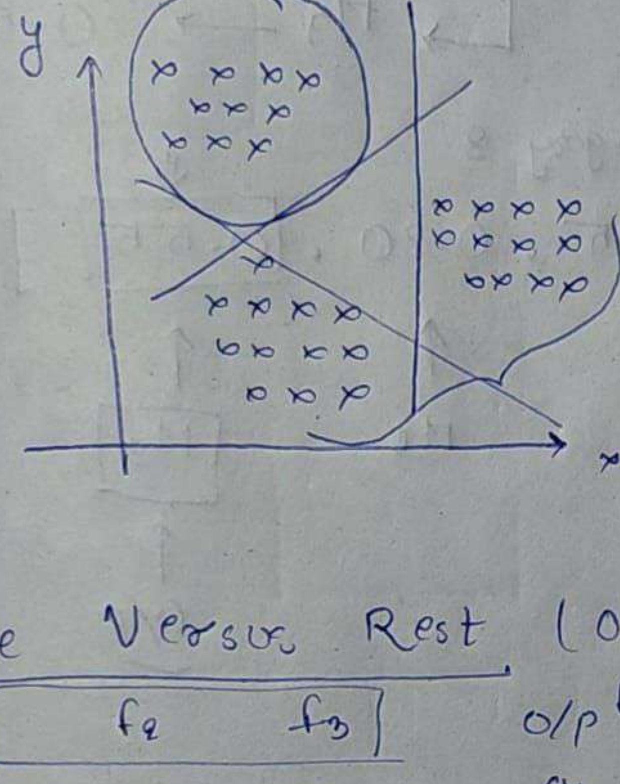
Logistic Regression (One Versus Rest)



(1) Binary Classification

One versus One



Multiclass classification {logistic Regression}

→ M1 → Binary classification
→ M2 → Binary classification
→ M3 → Binary classification

One Versus Rest (OVR) → logistic Regression

f1	f2	f3	o/p {O1, O2, O3}	O1	O2	O3
–	–	–	O1	1	0	0
–	–	–	O2	0	1	0
–	–	–	O3	0	0	1
–	–	–	O1	1	0	0
–	–	–	O3	0	0	1
–	–	–	O2	0	1	0

Model

↑

$M_1 \leftarrow$ I/P $\{f_1, f_2, f_3\}$ O/P $[\{O_1\}]$

$M_2 \leftarrow$ I/P $\{f_1, f_2, f_3\}$ O/P $[\{O_2\}]$

$M_3 \leftarrow$ I/P $\{f_1, f_2, f_3\}$ O/P $[\{O_3\}]$

New Test Data → $M_1 \rightarrow 0.25$

→ $M_2 \rightarrow 0.20$

→ $M_3 \rightarrow 0.55$

$0.55 \rightarrow$ O/P $\Rightarrow \boxed{O_3} \rightarrow$ category 3

$[0.25, 0.20, \boxed{0.55}]$

↑ M_1 ↑ M_2 ↑ $\boxed{M_3}$ ↑ O/P